

8. (a) The table shows information about some Group 1 elements. The melting point of rubidium is missing.

| Element   | Melting point (°C) | Boiling point (°C) | Density (g/cm <sup>3</sup> ) |
|-----------|--------------------|--------------------|------------------------------|
| lithium   | 181                | 1 330              | 0.53                         |
| sodium    | 98                 | 883                | 0.99                         |
| potassium | 64                 | 759                | 0.86                         |
| rubidium  |                    | 688                | 1.53                         |
| caesium   | 29                 | 671                | 1.93                         |

- (i) State which of the elements **lithium**, **sodium** or **potassium**, is a liquid over the greatest temperature range. [1]

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- (ii) Estimate a value for the melting point of rubidium. [1]

.....

- (iii) Group 1 elements are stored in paraffin oil to prevent reactions with oxygen.

The density of paraffin oil is 0.80 g/cm<sup>3</sup>.

State which of the Group 1 elements would float on paraffin oil. [1]

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- (b) When a small piece of **sodium** is added to water, it fizzes, melts and moves around on the surface of the water.

Give **another** observation that you would expect to make when **potassium** is added to water. State the reason for the difference. [2]

Observation .....

Reason .....

- (c) Lithium reacts with oxygen to give lithium oxide.

Complete the equation for the reaction by

- giving the formula for lithium oxide
- putting numbers in the boxes to balance the equation

[2]



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